

Annuities and Perpetuities — Worksheet

1. **Problem 1.** Tracy receives payments of $\$X$ at the end of each year for n years. The present value of her annuity is $\$493$. Gary receives payments of $\$3X$ at the end of each year for $2n$ years. The present value of his annuity is $\$2,748$. Both present values are calculated at the same annual effective interest rate. Find v^n .
2. **Problem 2.** Starting on his 25th birthday and continuing through his 60th birthday, Fred deposits $\$7,500$ each year on his birthday into a retirement fund earning an annual effective rate of 5%. Immediately after the last deposit, the accumulated value of the fund is transferred to a fund earning an annual effective rate of j . Five years later, a twenty-five year annuity-due paying $\$5,800$ each month is purchased with the funds. The purchase price of the annuity was determined using an annual effective rate of interest of 4%. Find j .
3. **Problem 3.** A scholarship fund pays a level scholarship of $\$X$ at the *end* of each year for 15 years. The fund's effective annual interest rate is 6% up until year 8 and then a discount rate of 4% after that. If the present value today of the 15 payments is $\$120,000$, find X .
4. **Problem 4.** The Browns wish to accumulate at least 150,000 at the time of their last deposit in a college fund for their daughter by contributing an amount A into the account at the end of each year for eighteen years. What is the smallest annual payment A that will suffice if the college fund earns a level annual effective interest rate of 5%? If at the end of ten years, it is announced that the annual effective interest rate will drop to 4.5%, how much must the Browns increase their payments in order to reach their accumulation goal? Assume that the Browns wish to continue to make level payments except for a slightly reduced final payment.
5. **Problem 5.** Suppose $\$40,000$ was invested on January 1, 1980 at an annual effective interest rate of 7% in order to provide an annual (calendar-year) scholarship of $\$5,000$ each year forever, the scholarships paid out each January 1.
 - (a) In what year can the first $\$5,000$ scholarship be made?
 - (b) What smaller scholarship can be awarded the year prior to the first $\$5,000$ scholarship?
6. **Problem 6.** An investor sets up a perpetuity-due that pays $\$10,000$ at the *beginning* of each year forever. For valuation, the effective annual discounting rate is 4% for payments in years 0–9 and 6% for payments in years ≥ 10 . Compute the present value today of this perpetuity-due.

Solutions

Notation / standard formulas used:

$$v = \frac{1}{1+i}, \quad a_{\overline{n}|i} = \sum_{t=1}^n v^t = \frac{1-v^n}{i}, \quad s_{\overline{n}|i} = \sum_{t=1}^n (1+i)^{t-1} = \frac{(1+i)^n - 1}{i},$$

$$\ddot{a}_{\overline{n}|i} = (1+i)a_{\overline{n}|i} = \frac{1-v^n}{d}, \quad d = \frac{i}{1+i}.$$

1. (Annuity Immediate, piecewise interest)

Derivation: A payments A occur at times $t = 1, 2, \dots, 10$ (end of years 1–10). Each payment made at time t accumulates to time 10 at rate 5% for $(10 - t)$ years, and then accrues at 4% for the remaining 10 years (from time 10 to 20). Therefore the accumulation to time 20 of the A -block is

$$A \cdot (1.04)^{10} \sum_{j=0}^9 1.05^j = A \cdot (1.04)^{10} \cdot \frac{1.05^{10} - 1}{0.05}.$$

The B -payments occur at $t = 11, 12, \dots, 20$ and each such payment accumulates to time 20 at 4% for the appropriate remaining years, so the B -block accumulates to

$$B \sum_{k=0}^9 1.04^k = B \cdot \frac{1.04^{10} - 1}{0.04}.$$

The required condition (total accumulation = \$2,000,000) is

$$A \cdot (1.04)^{10} \cdot \frac{1.05^{10} - 1}{0.05} + B \cdot \frac{1.04^{10} - 1}{0.04} = 2,000,000.$$

Define the two accumulation factors

$$F_A = (1.04)^{10} \cdot \frac{1.05^{10} - 1}{0.05}, \quad F_B = \frac{1.04^{10} - 1}{0.04}.$$

Numerically (to high precision)

$$F_A \approx 18.61835354206, \quad F_B \approx 12.00610712296.$$

Thus the linear relation between A and B is

$$A \cdot 18.61835354206 + B \cdot 12.00610712296 = 2,000,000,$$

or equivalently

$$B = \frac{2,000,000 - A \cdot 18.61835354206}{12.00610712296}.$$

This is the general solution family (one linear equation, two unknowns).

Special case: equal payments $A = B$. If we impose $A = B$ (a commonly useful special case), then

$$A(F_A + F_B) = 2,000,000 \quad \Rightarrow \quad A = \frac{2,000,000}{F_A + F_B}.$$

Numerically,

$$A = \frac{2,000,000}{18.61835354206 + 12.00610712296} \approx 65,307.27.$$

So if the corporation chooses equal payments for all 20 years, each payment would be \$65,307.27.

2. (Annuity Due — Janet)

Setup: Let time 0 be Janet's 41st birthday (first deposit). Deposits of size D are at times $0, 1, \dots, 19$ (20 beginning-of-year deposits). The annuity payments to Janet are \$30,000 at the beginning of years 61–80, which correspond to times $20, 21, \dots, 39$ relative to our time 0; there are 20 beginning payments (an annuity-due of 20 payments) starting at time 20.

Value required at time 20: The *present value at time 20* of the annuity-due of 20 payments of \$30,000 is

$$30,000 \cdot \ddot{a}_{\overline{20}|0.05} = 30,000 \cdot \frac{1 - (1.05)^{-20}}{d_{0.05}} \quad \text{where } d_{0.05} = \frac{0.05}{1.05}.$$

It is often convenient to use $\ddot{a}_{\overline{20}|0.05} = (1+i)a_{\overline{20}|0.05}$, so

$$\ddot{a}_{\overline{20}|0.05} = (1.05) \cdot \frac{1 - (1.05)^{-20}}{0.05}.$$

Accumulation of the deposits to time 20: Deposits D at times $0, 1, \dots, 19$ accumulate to time 20 to

$$D \cdot (1.05)^{s_{\overline{20}|0.05}},$$

because the accumulation of an annuity-due of n deposits to time n is $(1+i)s_n$.

Equation: Equate accumulation of deposits to the amount required at time 20:

$$D \cdot (1.05)^{s_{\overline{20}|0.05}} = 30,000 \cdot \ddot{a}_{\overline{20}|0.05}.$$

Hence

$$D = \frac{30,000 \cdot \ddot{a}_{\overline{20}|0.05}}{(1.05)^{s_{\overline{20}|0.05}}}.$$

Numeric evaluation: At $i = 0.05$,

$$\ddot{a}_{\overline{20}|0.05} \approx 13.0853208597, \quad s_{\overline{20}|0.05} \approx 33.0659541029.$$

Thus

$$D \approx \frac{30,000 \cdot 13.0853208597}{1.05 \cdot 33.0659541029} \approx 11,306.68.$$

$D \approx \$11,306.68$ (each beginning-of-year deposit).

3. (Annuity Immediate, piecewise interest — find level payment X)

Setup: Payments X at times $t = 1, \dots, 15$. Interest is 6% for years 1–8 and 4% for years 9–15.

Present value decomposition:

$$PV = X \sum_{t=1}^8 (1.06)^{-t} + X \sum_{t=9}^{15} \left((1.06)^{-8} (1.04)^{-(t-8)} \right).$$

The second sum is just $(1.06)^{-8} \sum_{s=1}^7 (1.04)^{-s}$ (where $s = t - 8$).

Therefore the PV factor (call it F) satisfying $PV = X \cdot F$ is

$$F = a_{\overline{8}|0.06} + (1.06)^{-8} \cdot a_{\overline{7}|0.04}.$$

Given: $PV = 120,000$. So $X = 120,000/F$.

Numeric evaluation:

$$a_{\overline{8}|0.06} \approx 6.20979381097, \quad a_{\overline{7}|0.04} \approx 6.00205466995,$$

$$F \approx 6.20979381097 + (1.06)^{-8} \cdot 6.00205466995 \approx 9.97555716437.$$

Hence

$$X \approx \frac{120,000}{9.97555716437} \approx 12,029.40.$$

$X \approx \$12,029.40$ per year (end of year payments).

4. (Annuity Due, piecewise interest — present value)

Setup: Payments \$5,000 at times $t = 0, 1, 2, \dots, 24$ (25 beginning payments). Interest is 7% for years 0–9 (affecting discount factors for times up to 9) and 4% for times ≥ 10 .

Present value calculation: Discount each payment back to time 0 using the piecewise structure.

$$PV = 5,000 \sum_{t=0}^9 (1.07)^{-t} + 5,000 \sum_{t=10}^{24} \left((1.07)^{-10} (1.04)^{-(t-10)} \right).$$

The second sum is $5,000(1.07)^{-10} \sum_{s=0}^{14} (1.04)^{-s}$.

Numeric evaluation:

$$\sum_{t=0}^9 (1.07)^{-t} \approx 7.51523224880, \quad (1.07)^{-10} \sum_{s=0}^{14} (1.04)^{-s} \approx 5.87810535606.$$

So

$$PV \approx 5,000 \cdot (7.51523224880 + 5.87810535606) \approx 66,966.69.$$

Purchase price $\approx \$66,966.69$.

5. (Perpetuity Immediate — equality condition)

Interpretation: The mathematics+engineering pair together receive \$100,000 at the end of each year for years 1–20, and \$0 thereafter. The medical school receives \$0 for years 1–20 and \$100,000

for years 21– ∞ . The donor wants the *present value at time 0* of (math+eng) to equal the PV of medical.

Write PVs:

$$PV_{M+E} = 100,000 \cdot a_{\overline{20}|i} \quad (\text{finite annuity of 20 payments}).$$

$$PV_{\text{Med}} = 100,000 \cdot \frac{v^{20}}{i} \quad (\text{deferred perpetuity: a perpetuity immediate deferred 20 years}).$$

Equate:

$$100,000 \cdot a_{\overline{20}|i} = 100,000 \cdot \frac{v^{20}}{i}.$$

Cancelling 100,000 and using $a_{\overline{20}|i} = (1 - v^{20})/i$ gives

$$\frac{1 - v^{20}}{i} = \frac{v^{20}}{i} \Rightarrow 1 - v^{20} = v^{20} \Rightarrow v^{20} = \frac{1}{2}.$$

Therefore

$$(1 + i)^{20} = 2 \Rightarrow i = 2^{1/20} - 1.$$

Numeric value:

$$i \approx 2^{1/20} - 1 \approx 0.03526492384 \approx 3.5265\%.$$

So the required annual effective interest rate is $i = 2^{1/20} - 1 \approx 3.5265\%$.

6. (Perpetuity Due, piecewise discounting)

Setup: Payments \$10,000 at times $t = 0, 1, 2, \dots$. For $t = 0, \dots, 9$ the appropriate effective discounting factor per year is 4% (so each of those payments is discounted with 1.04^{-t}). For $t \geq 10$ the discounting for the years ≥ 10 uses 6% per year (so to discount a payment at time $t \geq 10$ back to time 0, we multiply by $(1.04)^{-10}(1.06)^{-(t-10)}$).

PV: Split the perpetuity into the first 10 payments and the tail:

$$PV = 10,000 \sum_{t=0}^9 (1.04)^{-t} + 10,000 (1.04)^{-10} \sum_{s=0}^{\infty} (1.06)^{-s},$$

where $s = t - 10$ indexes the tail payments. The infinite geometric sum is

$$\sum_{s=0}^{\infty} (1.06)^{-s} = \frac{1}{1 - (1/1.06)} = \frac{1.06}{0.06}.$$

So

$$PV = 10,000 \left(\sum_{t=0}^9 (1.04)^{-t} + (1.04)^{-10} \cdot \frac{1.06}{0.06} \right).$$

Numeric evaluation:

$$\sum_{t=0}^9 (1.04)^{-t} \approx 8.43533161053, \quad (1.04)^{-10} \cdot \frac{1.06}{0.06} \approx 11.93496698259.$$

Hence

$$PV \approx 10,000 \cdot (8.43533161053 + 11.93496698259) \approx 203,702.99.$$

$$\boxed{PV \approx \$203,702.99.}$$

Notes / Remarks

- I have given both algebraic (symbolic) derivations and numerical evaluations. Each approximate monetary answer is rounded to the nearest cent in the boxed result.
- Problem 1 as stated admits infinitely many (A, B) pairs satisfying the single accumulation equation; I supplied the general linear relation and then the common special case $A = B$. If you'd like a different additional constraint (for example minimize total paid, or set B to some fixed fraction of A), tell me which constraint and I will compute the unique pair.
- If you prefer different notation (for example the $a_{\overline{n}}$ vs. $\ddot{a}_{\overline{n}}$ conventions, or using s_n explicitly), I can reformat the LaTeX to match your course style.